PRAKTIKUM BASIS DATA

MODUL 8

Scripting SQL – Scripting SQL Create

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JURNAL

SOAL 1

1. Buatlah desain database untuk suatu perusahaan: a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id. b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

```
-- ======================================== --
CREATE TABLE locations (
    location_id NUMBER PRIMARY KEY,
    city VARCHAR2(50)
);

-- ======================================== --
CREATE TABLE department (
    department_id NUMBER PRIMARY KEY,
    department_name VARCHAR2(50) NOT NULL,
    manager_id NUMBER,
    location_id NUMBER,
    CONSTRAINT fk_dept_location FOREIGN KEY (location_id)
REFERENCES locations(location_id)
);

-- ======================================== --
CREATE TABLE employee (
    employee_id NUMBER PRIMARY KEY,
    last_name VARCHAR2(50) NOT NULL,
    email VARCHAR2(100),
    salary NUMBER(10, 2),
    commission_pct NUMBER(4, 2),
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_emp_dept FOREIGN KEY (department_id)
REFERENCES department(department_id)
);
```

```
Script Output
Task completed in 0.081 seconds

Table LOCATIONS created.

Table DEPARTMENT created.

Table EMPLOYEE created.
```

2. Analisis table-table dengan compute statistics

```
Worksheet   Query Builder
    last_name VARCHAR2(50) NOT NULL,
    email VARCHAR2(100),
    salary NUMBER(10, 2),
    commission_pct NUMBER(4, 2),
    hire_date DATE NOT NULL,
    department_id NUMBER,
    CONSTRAINT fk_emp_dept FOREIGN KEY (department_id) REFERENCES department(department_id)
);

-- ===================================== --
EXEC DBMS_STATS.GATHER_TABLE_STATS(USER, 'LOCATIONS');
EXEC DBMS_STATS.GATHER_TABLE_STATS(USER, 'DEPARTMENT');
EXEC DBMS_STATS.GATHER_TABLE_STATS(USER, 'EMPLOYEE');

Script Output
Task completed in 0.635 seconds

Table EMPLOYEE created.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.
```

3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).

```
EXEC DBMS_STATS.GATHER_TABLE_STATS(USER, 'EMPLOYEE');

-- Insert Data
INSERT INTO locations VALUES (10, 'Jakarta');
INSERT INTO locations VALUES (20, 'Bandung');

INSERT INTO department VALUES (10, 'Administration', 100, 10);
INSERT INTO department VALUES (20, 'Marketing', 101, 20);
INSERT INTO department VALUES (80, 'Sales', 102, 10);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (1, 'King', 'king@mail.com', 5000, 0.1, SYSDATE, 80);
INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (2, 'Kochhar', 'kochhar@mail.com', 4500, NULL, SYSDATE, 80);
INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (3, 'De Haan', 'dehaan@mail.com', 4800, 0.2, SYSDATE, 10);
COMMIT;
```

Script Output — Task completed in 0.132 seconds

```
1 row inserted.

1 row inserted.

Commit complete.
```

4. Analisis table-table dengan compute statistics

```
    CONSTRAINT fk_emp_dept FOREIGN KEY (department_id) REFERENCES department(department_id)
);

-- ===================================== --
EXEC DBMS_STATS.GATHER_TABLE_STATS(USER, 'LOCATIONS');
EXEC DBMS_STATS.GATHER_TABLE_STATS(USER, 'DEPARTMENT');
EXEC DBMS_STATS.GATHER_TABLE_STATS(USER, 'EMPLOYEE');

-- Insert Data
INSERT INTO locations VALUES (10, 'Jakarta');
INSERT INTO locations VALUES (20, 'Bandung');

INSERT INTO department VALUES (10, 'Administration', 100, 10);
INSERT INTO department VALUES (20, 'Marketing', 101, 20);
INSERT INTO department VALUES (80, 'Sales', 102, 10);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
```

Script Output — Task completed in 0.463 seconds

```
PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.

PL/SQL procedure successfully completed.
```

5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.

```
-- ======================================== --

CREATE OR REPLACE VIEW empvu80 AS
SELECT employee_id AS id_number, last_name AS name, salary, department_id
FROM employee
WHERE department_id = 80;

-- ======================================== --
```

Script Output

Task completed in 0.054 seconds

```
PL/SQL procedure successfully completed.


PL/SQL procedure successfully completed.


View EMPVU80 created.
```

6. Kemudian tampilkan struktur dari view empvu80.

```
-- ======================================== --

DESC empvu80;
-- ======================================== --
```

Script Output

Task completed in 0.415 seconds

```
View EMPVU80 created.

Name          Null?    Type
------------- -------- ------------
ID_NUMBER     NOT NULL NUMBER
NAME          NOT NULL VARCHAR2(50)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER
```

7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

Worksheet | Query Builder

```
INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hi
VALUES (3, 'De Haan', 'dehaan@mail.com', 4800, 0.2, SYSDATE, 10);
COMMIT;

-- ======================================== --

CREATE OR REPLACE VIEW empvu80 AS
SELECT employee_id AS id_number, last_name AS name, salary, department_id
FROM employee
WHERE department_id = 80;

-- ======================================== --

DESC empvu80;
-- ======================================== --

CREATE INDEX emp_dept_id_idx ON employee(department_id);
```

Script Output

Task completed in 0.059 seconds

```
Name           Null?    Type
-------------- -------- ------------
ID_NUMBER      NOT NULL NUMBER
NAME           NOT NULL VARCHAR2(50)
SALARY                  NUMBER(10,2)
DEPARTMENT_ID           NUMBER

Index EMP_DEPT_ID_IDX created.
```